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Author(s): Kenneth Garbade

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Two Methods for Examining the Stability of Regression Coefficients

KENNETH GARBADE*

This paper investigates the power of two methodologies, the tests of Brown, Durbin, and Evans [2] and variable parameter regression, to detect several varieties of instability in the coefficients of a linear regression model. The study reported by Khan [10] on the stability of the demand for money is replicated with variable parameter regression, and his results are in part questioned and in part sharpened.

KEY WORDS: Kalman filter; Recursive residuals; Parameter stability; Cusum test; Cusum of squares test; Demand for money.

1. INTRODUCTION

In pursuing quantitative research, econometricians are frequently able to appeal to theoretical arguments to justify a choice for the functional form of an economic model. Rarely, however, are they able to justify a maintained hypothesis that the coefficients of a model are stable over a sample interval, for example, that the marginal propensity to consume was the same in the Great Depression as in the 1960s. Stability of estimated coefficients is an empirical question that has frequently been ignored. Among the early works in this field are Quandt [14] on estimating the location of a shift from one regression scheme to another, and Quandt [15] and Chow [4] on testing the null hypothesis of a shift at a particular point in time. Recently Brown, Durbin, and Evans [2] (hereafter BDE) proposed two more general methods of testing for the statistical significance of structural variation of regression coefficients, and Khan [10] applied one of those methods to test the stability of the demand for money in the United States.

The purpose of this paper is to investigate, using Monte Carlo simulation, the power of the BDE tests to reject a false null hypothesis of stable coefficients under a variety of regimes of coefficient variation. Phillips, Smith, and Harvey each suggested, in commenting on the BDE tests [2], that the usefulness of those tests depends on their behavior under specific alternative hypotheses. We will also compare the results of the BDE tests to those obtained from an application of variable parameter regression (VPR) [1, 16, 18] which can be used to measure the magnitude, as well as to test the significance, of coefficient variation.

The second section of the study describes variable pa-

rameter regression as it is applied here. The third section reviews the BDE tests and shows how they are related to a special case of the VPR model. The fourth section reports on Monte Carlo simulation experiments with the two techniques. We find that one of the BDE tests is generally less powerful than VPR in rejecting a false null hypothesis of stability, and that the power of the second test is, at least for the cases investigated here, quite weak. In the last section we replicate Khan's results [10] using VPR. We are led to question the hypothesis of stability in the demand for money in some cases, and to strengthen Khan's evidence for stability in others.

2. VARIABLE PARAMETER REGRESSION

Variable parameter regression assumes the researcher has observed T ordered scalar observations $(y_1, \dots, y_T)$ generated by the model:

$$y_t = x_t \beta_t + u_t, \quad t = 1, \dots, T, \quad (2.1)$$

$$u_t \sim N(0, \sigma^2) \quad (2.2)$$

where x_t is an n -dimensional row vector of known exogenous regressors and β_t is an n -dimensional column vector of unknown coefficients. The stochastic scalar u_t is drawn from a Gaussian distribution. The dynamic evolution of the β vector is assumed to follow a random walk with zero drift through time:

$$\beta_t = \beta_{t-1} + p_t, \quad t = 2, \dots, T, \quad (2.3)$$

$$p_t \sim N(0, \sigma^2 P) \quad (2.4)$$

where $\sigma^2 P$ is the stationary covariance matrix of the innovation p_t . If $P = 0$, then VPR collapses to the ordinary stationary coefficient linear regression problem. The normalizing variance term σ^2 of (2.4) is the same as that which appears in (2.2) and is shown explicitly for expositional convenience. The innovations u_t and p_t are mutually and serially uncorrelated. Although more complex models of the dynamic evolution of the β 's are possible [16, 18] we retain the simple form of (2.3) here. We are not concerned with estimating a particular stochastic process generating the β 's, but rather with the stability of the β 's and with the pattern of any indicated variation. If a researcher is led to reject the hypothesis of stable

*Kenneth Garbade is Assistant Professor, Departments of Economics and Finance, Graduate School of Business Administration, New York University, New York, NY 10006. This work was undertaken while the author was Research Associate at Princeton University in the summer of 1975. The author wishes to thank Gregory Chow, Adrian Pagan, Richard Quandt, and the referees for helpful suggestions.

coefficients, he or she might wish to consider structural explanations of the β 's as functions of additional exogenous variables rather than develop more complicated models of stochastically evolving coefficients. For example, the marginal propensity to consume could be a function of wealth or consumer sentiment. Results in Section 4 indicate the simple model will be adequate in many cases.

Given the observations $(y_1, \dots, y_T)$, the researcher is interested in estimating the trajectory of the β_t coefficients. Assume for the moment P and σ^2 are known. Let $\hat{\beta}_t(s)$ be the estimate of β_t based on the observations $(y_1, \dots, y_s)$, let $\sigma^2 R_t$ be the estimation covariance matrix of $\hat{\beta}_t(t-1)$ and let $\sigma^2 S_t$ be the estimation covariance matrix of $\hat{\beta}_t(t)$. The prediction step of VPR updates $\hat{\beta}_{t-1}(t-1)$ and its covariance for the passage of time from period $t-1$ to period t :

$$\hat{\beta}_t(t-1) = \hat{\beta}_{t-1}(t-1), \quad (2.5)$$

$$R_t = S_{t-1} + P. \quad (2.6)$$

The filter step revises this estimate of β_t by adding the new information contained in the observation y_t :

$$\hat{\beta}_t(t) = \hat{\beta}_t(t-1) + K_t(y_t - x_t \hat{\beta}_t(t-1)) \quad (2.7)$$

where $K_t = R_t x_t' [x_t R_t x_t' + 1]^{-1}$,

$$S_t = R_t - K_t x_t R_t. \quad (2.8)$$

K_t is the so-called gain matrix. When all the filtered estimates are available, smoothing adds the information which arrived after time t to the estimate of β_t . Let $\sigma^2 V_t$ be the estimation covariance of the smoothed estimate $\hat{\beta}_t(T)$. The smoothed estimates are computed as:

$$\hat{\beta}_t(T) = \hat{\beta}_t(t) + G_t[\hat{\beta}_{t+1}(T) - \hat{\beta}_t(t)] \quad (2.9)$$

where $G_t = S_t[S_t + P]^{-1}$,

$$V_t = S_t + G_t[V_{t+1} - R_{t+1}]G_t', \quad (2.10)$$

$$V_T = S_T. \quad (2.11)$$

If $P = 0$ it can be shown that $\hat{\beta}_t(T)$ and V_t are constant for all t and equal to their ordinary least-squares analogues. These results are derived at length in [17]. The scalar stochastic innovation $e_t = y_t - x_t \hat{\beta}_t(t-1)$ driving the filter is a zero mean, serially uncorrelated Gaussian process with variance $\sigma^2 E_t$, where $E_t = 1 + x_t' R_t x_t$. The properties of the innovations are conveniently described in [12].

In most cases the researcher will not have a prior estimate of β_1 , of the form $(\hat{\beta}_1(0), \sigma^2 R_1)$, with which to begin the recursive estimation at the filter step. Garbade [8] has suggested the algorithm can be initialized by using the first n observations to estimate β_{n+1} . In this case we have:

$$\hat{\beta}_{n+1}(n) = M \left\{ \sum_{t=1}^n H_t' x_t' e_t^* / E_t^* \right\}, \quad (2.12)$$

$$R_{n+1} = S_n^* + P + H_{n+1} M H_{n+1}' \quad (2.13)$$

where

$$M = \left\{ \sum_{t=1}^n H_t' x_t' x_t H_t / E_t^* \right\}^{-1}, \quad (2.14)$$

$$H_t = (I - K_{t-1} x_{t-1}) H_{t-1}, \quad t = 2, \dots, n \quad (2.15)$$

with $H_1 = I$. E_t^* , e_t^* , and S_t^* are computed from (2.5) to (2.8) for the special initial conditions $\hat{\beta}_1(0) = 0$, $R_1 = 0$.

The relative covariance matrix P and the variance scalar σ^2 can be estimated by maximum likelihood. Schweppe [19] has shown the log-likelihood function of P and σ^2 , to within a constant, is

$$L = -\frac{1}{2} \sum_{t=n+1}^T \ln(\sigma^2 E_t) - \frac{1}{2} \sum_{t=n+1}^T \{e_t^2 / (\sigma^2 E_t)\}. \quad (2.16)$$

An estimate of σ^2 , for given P , is available by analytic maximization of (2.16):

$$\hat{\sigma}^2 = \sum_{t=n+1}^T \{e_t^2 / ((T-n)E_t)\}. \quad (2.17)$$

The concentrated log-likelihood function, to within a constant, is then:

$$L^*(P) = -(T-n) \ln(\hat{\sigma}) - \frac{1}{2} \sum_{t=n+1}^T \ln(E_t) \quad (2.18)$$

where $\hat{\sigma}$ and E_t are both implicit functions of P . L^* is thus a complicated nonlinear function of P . Numerical methods must be used to maximize L^* as a function of the unknown elements in the matrix P . For a sample of fixed length T , L^* is a continuous function of P in an open neighborhood of the closed set of symmetric $n \times n$ nonnegative definite matrices. In particular, it is a continuous function of P in an open set around $P = 0$. For example, if P is of dimension one, $L^*(P)$ is a continuous function for all $P > -\epsilon$ for some $\epsilon > 0$. As long as $E_t > 0$ for all t , (2.16) shows that L^* will be defined. This condition on the E_t 's can be satisfied for small negative values of P as well as for $P \geq 0$ (cnf. (2.6) and (2.8)). Since estimates of P that are not nonnegative definite are meaningless, maximization of the likelihood function should be restricted to the set of nonnegative definite matrices. In the case of $n = 1$ this means maximizing L^* on the nonnegative half-line. It follows that, in some cases, the first-order condition for a maximum of the concentrated likelihood function will not hold. As the sample length increases, the value of ϵ such that $L^*(P)$ is defined for all $P > -\epsilon$ goes to zero and this problem becomes less important.

Whether the maximum likelihood estimate $\hat{P}$ is significantly different from an hypothesized value P_0 can, in some circumstances, be tested with the likelihood statistic:

$$-2 \ln(\lambda) = -2[L^*(P_0) - L^*(\hat{P})]. \quad (2.19)$$

If the parameter vector has dimension $n = 1$, so that P is a scalar, the model of (2.1) to (2.4) may be viewed as a reparameterization of the model considered by Cooley and Prescott [5]. It follows from the results of those

authors and Kendall and Stuart [9, p. 240] that, for $P_0 > 0$, the likelihood statistic (2.19) will (asymptotically) have a chi-square distribution with one degree of freedom under a true null hypothesis. For $P = P_0 = 0$, however, the likelihood statistic will be more concentrated toward the origin than a chi-square distribution because maximum likelihood estimates less than zero are set to zero. This is clearly evident in the empirical distributions shown in Table 1. Using a chi-square distribution to determine the critical values of $-2 \ln(\lambda)$ will thus lead to a conservative test of the stability of the β_t coefficients.

For $n > 1$ the likelihood statistic of (2.19) can be used to test the null hypothesis that particular diagonal terms in P are zero under the maintained hypothesis that the remaining $n - 1$ diagonal elements are zero. This is also a reparameterization of the model of Cooley and Prescott. The statistic $-2 \ln(\lambda)$ will again be more concentrated than a chi-square distribution with one degree of freedom under a true null hypothesis.

The distribution of the likelihood statistic in the general case (where all the elements of P are estimated) under a true null hypothesis of $P_0 = 0$ will not, in general, be related to a chi-square distribution. Consider the case where only the diagonal elements of P are to be estimated. If the off-diagonal elements of $R = [\text{Exp} [d^2 L^*(0)/(dP)^2]]^{-1}$ were known to be zero, one might be able to appeal to an independence argument on the estimates of the elements of P and assume $-2 \ln(\lambda)$ is more concentrated than a chi-square distribution with n degrees of freedom. This follows from the symmetry of the normal distribution. Outside of special cases, however, R is unlikely to be a diagonal matrix, and the distribution of $-2 \ln(\lambda)$ consequently unknown.

In addition to formal hypothesis testing, VPR provides the researcher with a graphical presentation of the estimated evolving coefficients based on the full set of observations, i.e., with the $\hat{\beta}_t(T)$'s, and with standard errors for those estimates, the square roots of the diagonal elements of $\sigma^2 V_t$. These are useful in deciding whether the variation of the coefficients is quantitatively important as well as statistically significant. They may also provide guidance if the coefficients were shifting over part of the sample interval but were stable over the remainder.¹ One of the simulations reported in Section 4 will address the question of the usefulness of VPR when the coefficients shift infrequently.

3. THE BROWN, DURBIN, AND EVANS TESTS

Both BDE tests begin with the computation of a series of recursive residuals. These residuals are closely related to the e_t innovations in the VPR algorithm. Consider (2.1). Let X_t' be the $n \times t$ matrix $(x_1', \dots, x_t')$ and let Y_t' be

the t -dimensional row vector $(y_1, \dots, y_t)$ for $n \leq t \leq T$. Under the null hypothesis that the coefficients in (2.1) are stable, the least-squares estimate of β based on observations to time t is $b_t = (X_t' X_t)^{-1} X_t' Y_t$ with estimation covariance matrix $\sigma^2 (X_t' X_t)^{-1}$. BDE define the recursive scalar-valued residuals w_t ,

$$w_t = \frac{y_t - x_t b_{t-1}}{[1 + x_t (X_{t-1}' X_{t-1})^{-1} x_t']^{\frac{1}{2}}}, \quad t = n+1, \dots, T, \quad (3.1)$$

and show that, under the null hypothesis, they form a serially independent Gaussian process with mean zero and variance σ^2 .

Actually, the recursive residuals of BDE are exactly the e_t innovations of VPR (divided by a scaling factor to make their variance equal to σ^2) when the hypothesis of $P = 0$, i.e., stable coefficients, is maintained.² Moreover, the recursive algorithm suggested by BDE to compute efficiently the b_t 's is identical to the prediction and filtering components of VPR when P is set to zero. Thus the BDE tests and the VPR algorithm are closely related under a true null hypothesis of $P = 0$.

Brown, Durbin, and Evans suggest two tests of the stability of regression coefficients, both of which use the recursive residuals. In the "cusum" test the cumulative sum series,

$$W_t = \sum_{s=n+1}^t w_s / \hat{\sigma}, \quad t = n+1, \dots, T, \quad (3.2)$$

where $\hat{\sigma}^2 = (Y_T - X_T b_T)'(Y_T - X_T b_T)/(T - n)$ is computed.³ Under the null hypothesis of stability W_t is approximately normally distributed with mean zero and variance $t - n$. The null hypothesis is rejected if $|W_t| > [a(T - n)^{\frac{1}{2}} + 2a(t - n)(T - n)^{-\frac{1}{2}}]$ for any $t \in [n+1, T]$, where the scalar a is chosen to obtain the desired confidence level. BDE suggest $a = 1.143$ for a 1 percent confidence level, and $a = .948$ for a 5 percent level. These values of a , however, are exact only for a continuous Brownian process with Gaussian noise. Anderson pointed out [2] that the true probability of rejection for a periodically sampled system is exaggerated by the suggested confidence levels. This is evident in our simulation results of Section 4.

In the "cusum of squares" test the cumulative sum of squares series,

$$S_t = \sum_{s=n+1}^t w_s^2 / \sum_{s=n+1}^T w_s^2, \quad t = n+1, \dots, T, \quad (3.3)$$

is computed. S_t is a monotonically increasing sequence of positive numbers with $S_T = 1$. Under the null hypothesis of stability $1 - S_t$ has a beta distribution⁴ with parame-

² This was noted by Priestly and Young in discussing the BDE tests [2].

³ Harvey [2] suggests that if $\hat{\sigma}^2$ is defined as $\hat{\sigma}^2 = \sum (w_t - \bar{w})^2 / (T - n - 1)$, where $\bar{w} = \sum w_t / (T - n)$, the cusum test will likely be more powerful under certain alternative hypotheses. This alternative definition of $\hat{\sigma}^2$ does not alter the distribution of W_t under the null hypothesis.

⁴ The distribution of S_t may be derived by noting that,

$$S_t^{-1} - 1 = \sum_{s=t+1}^T w_s^2 / \sum_{s=n+1}^t w_s^2.$$

¹ In discussing the BDE tests, Young [2] pointed out the usefulness of the recursive or filtered coefficient estimates in judging the nature of an instability. The filtered estimates, however, do not use forward observations and are typically quite erratic in the early periods of a sample interval. The smoothed estimates, computed from the estimated value of P , may be more informative.

ters $\alpha = -1 + (T - n)/2$ and $\beta = -1 + (t - n)/2$ and therefore, S_t has mean value $(t - n)/(T - n)$. BDE suggest constructing a confidence interval for S_t as $[(t - n)/(T - n)] \pm c_0$, where c_0 is chosen from Table 1 of [6] using $\alpha = .5$ times the confidence level desired and $n = \frac{1}{2}(T - n) - 1$ in that table. If $|S_t - ((t - n)/(T - n))| > c_0$ for any $t \in [n + 1, T]$ the null hypothesis is rejected.

4. SIMULATION RESULTS

In this section we present Monte Carlo simulation results obtained from applying vPR and the BDE tests to three different patterns of variation in the coefficients of a linear regression model. The patterns were chosen to provide one where the BDE tests are expected to be quite powerful, one where vPR is appropriate, and one where neither is appropriate. A true null hypothesis of stability is investigated as a special case of one of the patterns. The objective of these simulations is to get some idea of the relative merits of the two methodologies under various regimes of coefficient variation.

We found the likelihood test from vPR, based on the statistic of (2.19), had the desirable property of becoming increasingly powerful in rejecting a false null hypothesis of coefficient stability as a function of both the length of the sample interval and the magnitude of the coefficient instability. The power of the BDE cusum of squares test increased as a function of the magnitude of the instability, but did not grow monotonically more powerful as the sample interval increased. With few exceptions the vPR test was more powerful than the cusum of squares test. The cusum test was quite weak in every case.

We felt the most interesting results could be obtained from investigating a simple model in a variety of circumstances. The observation model we investigated was:

$$y_t = x_t \beta_t + u_t, \quad t = 1, \dots, T, \quad (4.1)$$

$$u_t \sim N(0, 1) \quad (4.2)$$

where β_t and x_t are both scalars. Note that σ^2 is set to unity. The sample length T was set variously to 15, 31, and 61 observations. The values of x_t were chosen randomly from a normal distribution with mean zero and variance 25. The value of x_t for any given t was chosen once and remained identical across all replications and patterns of coefficient variation.

We studied three patterns of coefficient behavior:

Case I: Random Walk with Zero Drift

$$\beta_t = \beta_{t-1} + p_t, \quad t = 2, \dots, T, \quad (4.3)$$

$$\beta_1 = 1, \quad (4.4)$$

$$p_t \sim N(0, P).$$

Case II: Discrete Jump

$$\begin{aligned} \beta_t &= 1, & 1 \leq t < \frac{1}{2}(T - 1), \\ &= 1 + \Delta, & \frac{1}{2}(T - 1) \leq t \leq T. \end{aligned} \quad (4.5)$$

Case III: Stable Markov Process

$$\beta_t = .7 + .3\beta_{t-1} + p_t, \quad t = 2, \dots, T, \quad (4.6)$$

$$\beta_1 = 1, \quad (4.7)$$

$$p_t \sim N(0, P).$$

A true null hypothesis of stable coefficients was studied as a special instance of Case I by setting P equal to zero. Simulation results are presented for Cases I and III when $P = .01, .1$, and 1 , and for Case II when $\Delta = \delta, 10\delta$, and 100δ , where $\delta = (25T)^{-\frac{1}{2}}$. The coefficient shift Δ was scaled by the relative shift δ to reflect the length of the sample set. The β_t coefficients in Cases I and III were redrawn in every replication. The statistics reported for each experiment were obtained from 300 replications.

Case II, where the coefficient changes by a discrete amount, should favor the BDE tests. The discrete jump case seriously violates the assumptions of the vPR model, since that model assumes an identically distributed increment to the previous value of the coefficient in every period. Case I is the vPR model of (2.3). Case III is similar to I but the coefficients revert toward unity instead of following a random walk. Cases I and III are both special cases of the more general model,

$$\Delta\beta_t = M(\beta^* - \beta_{t-1}) + p_t \quad (4.8)$$

where M is an adjustment matrix whose eigenvalues are nonnegative and not greater than two. If M equals the identity matrix then $\beta_t = \beta^* + p_t$ and we have a stochastic coefficient regime. Knowledge of β_{t-1} is of no value in predicting β_t , and the stochastic innovation p_t has no persistent effect on subsequent β 's. If M equals the zero matrix we have Case I, where the effect of the p_t innovations are fully persistent. Case III is between these two extremes. There is some persistence in the effect of p_t on subsequent values of the β 's, but the magnitude of the effect diminishes as time goes on.⁵

The first simulation results concern the empirical distribution of the likelihood statistic (2.19) under a true null hypothesis for P_0 . Table 1 shows the distribution of that statistic from the vPR model (Case I) for four values of P , including $P = 0$, when the hypothesized value of P_0 in (2.19) equals the true value of P . The width of the cells in Table 1 were chosen so that a chi-square variate with one degree of freedom has equal probability of appearing in any of the cells. The number of cells was set according to the suggestion of Mann and Wald [11] (see also [21]). When $P = 0$ the likelihood statistic is heavily concentrated near zero. The hypothesis that this statistic has a chi-square distribution with one degree of freedom was easily rejected with a chi-square test. For the three

Since the numerator and denominator of $S_t^{-1} - 1$ are sums of squares of independently distributed normal variates with zero means and equal variances it follows that $(S_t^{-1} - 1)(t - n)/(T - t)$ is distributed as an F statistic with $(T - t, t - n)$ degrees of freedom [13]. The result for S_t is then immediate since if the statistic v has an F distribution with (m, n) degrees of freedom then $w = mv/(n(1 + mv/n))$ has a beta distribution with parameters $\alpha = (m - 2)/2$ and $\beta = (n - 2)/2$ [13].

⁵ Rosenberg [16] treats (4.8) in some detail. We remind the reader that our purpose here is not the estimation of the parameters of (4.8) but rather the use of the simpler model (2.3) in detecting coefficient instability.

nonzero values of P , however, the hypothesis that the statistic has a chi-square distribution could not be rejected even at the 50 percent confidence level, despite the fact that the sample length, $T = 31$, is not large. Table 1 supports the statements of Section 2 that a chi-square test on the significance of $-2 \ln(\lambda)$ is asymptotically valid for $P_0 > 0$ and is conservative for $P_0 = 0$.

1. Distribution of Likelihood Statistic Under True Null Hypothesis on P in VPR Model^a

Cell	Upper boundary of cell	P			
		.0	.01	.1	1.0
1	.00136	.643	.017	.037	.037
2	.00545	.020	.040	.027	.060
3	.0123	.007	.020	.030	.020
4	.0219	.010	.027	.043	.013
5	.0344	.017	.013	.027	.040
6	.0497	.003	.037	.027	.017
7	.0681	.007	.037	.043	.020
8	.0896	.007	.023	.017	.043
9	.1143	.010	.027	.037	.020
10	.1424	.013	.033	.043	.013
11	.1741	.007	.047	.013	.053
12	.2097	.003	.027	.037	.033
13	.2492	.013	.027	.027	.030
14	.2931	.010	.037	.030	.030
15	.3418	.003	.030	.027	.040
16	.3956	.017	.037	.020	.030
17	.4550	.020	.040	.020	.033
18	.5206	.007	.020	.030	.023
19	.5933	.010	.023	.017	.023
20	.6738	.007	.020	.027	.023
21	.7632	.020	.030	.040	.020
22	.8630	.017	.027	.033	.030
23	.9748	.017	.030	.020	.027
24	1.1009	.003	.040	.030	.037
25	1.2442	.017	.030	.040	.027
26	1.4090	.007	.023	.027	.023
27	1.6007	.017	.043	.010	.020
28	1.8272	.020	.010	.030	.037
29	2.1027	.003	.017	.033	.033
30	2.4488	.023	.013	.020	.033
31	2.9074	.010	.027	.053	.023
32	3.5711	.033	.057	.020	.033
33	4.7464	.010	.043	.037	.033
34	∞	.0	.030	.030	.020

^a Based on 300 replications of a sample of length $T = 31$. Columns may not sum to unity due to rounding.

Turning to the question of the power of the two methodologies to reject a false null hypothesis of stability, Table 2a shows the rejection incidence of the null hypothesis for the VPR likelihood test and the BDE tests when the coefficients in (4.1) evolve according to Case I. The critical values of the likelihood statistic were taken from a chi-square distribution with one degree of freedom, although the results cited in the previous paragraph showed those choices overstate the probability of rejecting a true null hypothesis of stability. Table 2a indicates that the power of the VPR likelihood test in rejecting a false null hypothesis increases monotonically with both P and T . Larger values of P may be interpreted as implying more unstable coefficients. The power of the cusum of squares test does not increase monotonically with T , and

2. Incidence of Rejection of Null Hypothesis of Stability at 5 Percent and 1 Percent Level of Confidence when True Model Is a Random Walk, a Discrete Jump, or a Stable Markov Process^a

Test	Sample length					
	$T = 15$		$T = 31$		$T = 61$	
	5%	1%	5%	1%	5%	1%
a. Random walk						
$P = 0$						
BDE Cusum	1.0	.0	2.7	.3	3.0	.7
BDE Cusum of Squares	5.7	.3	4.7	.7	6.3	.7
VPR Likelihood	.7	.0	.7	.0	1.3	.3
$P = .01$						
BDE Cusum	.0	.0	2.7	.0	1.3	.0
BDE Cusum of Squares	25.3	12.3	20.7	8.3	40.7	28.3
VPR Likelihood	36.3	21.7	63.7	47.7	91.7	84.3
$P = .1$						
BDE Cusum	.3	.0	1.0	.0	1.3	.0
BDE Cusum of Squares	56.3	42.3	54.0	30.7	73.7	59.0
VPR Likelihood	81.3	74.3	98.3	95.7	100.0	100.0
$P = 1.$						
BDE Cusum	.0	.0	.3	.0	.7	.0
BDE Cusum of Squares	62.7	46.3	70.3	43.0	77.7	65.7
VPR Likelihood	96.3	92.7	100.0	100.0	100.0	100.0
b. Discrete jump						
$\Delta = \delta$						
BDE Cusum	1.0	.0	2.0	.0	4.0	.0
BDE Cusum of Squares	6.3	.0	6.7	1.3	7.7	2.2
VPR Likelihood	2.3	.0	1.3	.0	2.3	.0
$\Delta = 10\delta$						
BDE Cusum	4.3	.0	.7	.0	2.7	.0
BDE Cusum of Squares	3.3	.0	7.0	.7	14.0	3.3
VPR Likelihood	65.0	38.0	79.7	58.0	88.0	73.0
$\Delta = 100\delta$						
BDE Cusum	.0	.0	.0	.0	.0	.0
BDE Cusum of Squares	.0	.0	100.0	100.0	100.0	100.0
VPR Likelihood	100.0	100.0	100.0	100.0	100.0	100.0
c. Stable Markov process						
$P = .01$						
BDE Cusum	.3	.0	3.3	.3	4.0	1.0
BDE Cusum of Squares	7.7	2.0	7.7	2.3	8.0	1.7
VPR Likelihood	7.3	1.3	13.3	4.7	27.3	14.7
$P = .1$						
BDE Cusum	1.3	.0	2.3	.0	1.7	.3
BDE Cusum of Squares	34.0	17.0	23.3	10.7	34.3	17.0
VPR Likelihood	46.7	30.0	74.3	60.7	93.3	89.0
$P = 1.$						
BDE Cusum	.7	.0	4.3	1.0	3.3	.7
BDE Cusum of Squares	54.3	33.0	41.7	19.0	56.7	34.4
VPR Likelihood	83.0	73.0	97.7	94.7	100.0	100.0

^a Entries are in percent.

3. Parameter Estimates when True Model Is a
Random Walk, a Discrete Jump, or a
Stable Markov Process^a

Parameter	Sample length		
	T = 15	T = 31	T = 61
<i>a. Random walk</i>			
<i>P = 0</i>			
Stationary σ^2	1.02 (.40)	.98 (.25)	.99 (.18)
VPR $P\sigma^2$	.0021 (.0122)	.0005 (.0017)	.0001 (.0003)
VPR σ^2	.95 (.40)	.94 (.24)	.98 (.18)
<i>P = .01</i>			
Stationary σ^2	1.70 (.94)	1.97 (1.02)	2.80 (1.85)
VPR $P\sigma^2$	.0134 (.0250)	.0098 (.0091)	.0098 (.0062)
VPR σ^2	.97 (.45)	1.03 (.31)	1.02 (.21)
<i>P = .1</i>			
Stationary σ^2	7.95 (7.78)	10.74 (9.24)	18.97 (18.00)
VPR $P\sigma^2$	.105 (.091)	.102 (.057)	.102 (.036)
VPR σ^2	1.04 (.65)	1.04 (.43)	1.00 (.26)
<i>P = 1.0</i>			
Stationary σ^2	67.99 (78.63)	98.88 (91.52)	180.89 (179.53)
VPR $P\sigma^2$	1.02 (.57)	1.01 (.37)	1.01 (.26)
VPR σ^2	1.12 (1.23)	1.06 (.69)	.99 (.40)
<i>b. Discrete jump</i>			
<i>$\Delta = \delta$</i>			
Stationary σ^2	.99 (.36)	.98 (.24)	1.00 (.17)
VPR $P\sigma^2$	.0020 (.0074)	.0005 (.0018)	.0001 (.0003)
VPR σ^2	.92 (.38)	.94 (.24)	.98 (.17)
<i>$\Delta = 10\delta$</i>			
Stationary σ^2	2.17 (.63)	1.51 (.36)	1.31 (.22)
VPR $P\sigma^2$	.0333 (.044)	.0056 (.0060)	.0016 (.0012)
VPR σ^2	.99 (.50)	.94 (.26)	.98 (.18)
<i>$\Delta = 100\delta$</i>			
Stationary σ^2	143.26 (5.99)	56.94 (2.86)	31.50 (1.45)
VPR $P\sigma^2$	1.457 (.27)	.382 (.082)	.087 (.013)
VPR σ^2	1.05 (.78)	.72 (.38)	.85 (.21)

Table 3 (Continued)

Parameter	Sample length		
	T = 15	T = 31	T = 61
<i>c. Stable Markov process</i>			
<i>P = .01</i>			
Stationary σ^2	1.20 (.46)	1.27 (.32)	1.31 (.24)
VPR $P\sigma^2$	.0051 (.0145)	.0033 (.0061)	.0023 (.0037)
VPR σ^2	1.01 (.43)	1.10 (.33)	1.13 (.24)
<i>P = .1</i>			
Stationary σ^2	3.23 (1.86)	3.35 (1.28)	3.89 (1.38)
VPR $P\sigma^2$	.071 (.092)	.061 (.054)	.065 (.040)
VPR σ^2	1.26 (.80)	1.30 (.60)	1.22 (.44)
<i>P = 1.0</i>			
Stationary σ^2	23.48 (16.57)	24.22 (11.83)	29.89 (11.40)
VPR $P\sigma^2$	.937 (.674)	.946 (.471)	.970 (.323)
VPR σ^2	1.77 (2.20)	1.51 (1.25)	1.26 (.57)

^a Sample mean from 300 replications on top line, sample standard deviation underneath in parentheses.

is dominated in every case by the VPR likelihood test. The cusum test is quite weak in rejecting the null hypothesis.⁶ From inspection of the case $P = 0$ we see that the confidence levels on the cusum statistic overstate the actual probabilities of rejecting a true null hypothesis. As Anderson [2] pointed out, this is a consequence of applying the continuous Brownian approximation to a discretely sampled system.

Table 2b shows the rejection incidence for the three tests in the case where the coefficients change in value but once by a discrete amount. Except for the case of $\Delta = \delta$, the VPR likelihood test dominates the BDE cusum of squares test. The cusum test is, again, quite weak.

Table 2c shows the rejection incidence for the three tests when the coefficients follow the stable Markov process of Case III. It is interesting to compare the incidences of rejection for this pattern of incomplete persistence to those for the Case I pattern of complete persistence in changes in the values of the coefficients. The power to reject a false null hypothesis of both the VPR likelihood test and the BDE cusum of squares test is reduced when changes in the coefficients are incompletely persistent. This is clear from comparison of the corre-

⁶ Farley, Hinich, and McGuire [7] have pointed out that the BDE tests have poor asymptotic properties as the number of observations increases in a regime of coefficient variation different from those studied here. In particular, the cusum of squares test does not have an asymptotic power of 1 and the cusum test is inconsistent when the coefficient begins to change linearly with time at some point during the sample interval.

sponding entries of Tables 2a and 2c. The cusum of squares test, however, appears to suffer a greater reduction in power relative to the likelihood test.

Table 3 reports the sample means and sample standard deviations of the estimated observation error variance σ^2 for the stationary regression model, the estimated observation error variance σ^2 for the VPR model, and the estimated coefficient variance $P\sigma^2$ for the VPR model. When the VPR model is valid we see, in Table 3a, that the bias in the estimates of $P\sigma^2$ and σ^2 from VPR is quite small and that both the bias and sample standard deviation decrease as the sample interval T increases.

The statistics of Table 3a also suggest why the cusum test did so poorly in detecting the coefficient instability of Case I. The mean estimated value of σ^2 from the stationary coefficient regression model grows monotonically with both P and T . For Case I the observation model may be rewritten:

$$y_t = x_t\beta_1 + u_t + x_t \sum_{s=2}^t p_s. \quad (4.9)$$

When the coefficient follows a random walk with zero drift, the recursive residual w_t has the same expected value of zero which it has under a true null hypothesis. The cusum statistic W_t is thus not biased by the coefficient instability, so the probability that it will exceed a critical level depends upon its variance. This variance, it will be noted, depends inversely on the magnitude of δ . As (4.9) shows, δ depends upon the value of P . At least for the cases studied here, the variance of W_t is a decreasing function of P . Similar comments apply to the distribution of the cusum statistic under the coefficient variation pattern of Case III.

In Case II, where the coefficient changed by a discrete amount midway through the sample, the recursive residual is biased away from zero after the change. However, the bias is positive if $x_t > 0$, and negative if $x_t < 0$. Since x_t is a serially uncorrelated process, we rarely observed a sustained growth of the cusum statistic away from zero sufficient for that statistic to exceed a critical value. If the x_t series had been serially correlated, the cusum statistic would doubtless have displayed more power in rejecting a false null hypothesis when β changed by a discrete amount.

Table 3c shows that when the coefficient dynamics of (2.3) are assumed in a case where those of (4.6) are correct, the variance σ^2 of the transitory noise will be overstated and the coefficient variance $P\sigma^2$ underestimated. This occurs because β_t has a stable distribution around a mean value of unity in the Case III experiments. The mean estimate of $P\sigma^2$ from VPR will be less than the true $P\sigma^2$ of the stable coefficient dynamics, reflecting the structure of incomplete persistence in those dynamics.

Table 4 shows the sample mean and sample standard deviation of the standardized errors of the smoothed coefficient estimates for an illustrative example from each pattern of coefficient variation. The standardized estima-

tion errors were computed in each replication as:

$$z_t = (\hat{\beta}_t(T) - \beta_t)/(\delta V_t^{1/2}), \quad t = 2, \dots, T, \quad (4.10)$$

where $\hat{\beta}_t(T)$ and V_t are predicated on the estimated $\hat{P}$ from the given replication. The estimate $\hat{P}$ was used regardless of whether the likelihood test indicated that it was significantly different from zero. The entries of Table 4 were computed as:

$$\bar{z}_t = \sum_{r=1}^{300} z_t(r)/300, \quad (4.11)$$

$$s_t^2 = \sum_{r=1}^{300} (z_t(r) - \bar{z}_t)^2/299 \quad (4.12)$$

where $z_t(r)$ is the standardized error at time t in the r th replication. If the VPR model is true and $\hat{P} = P$, $\bar{z}_t$ is normally distributed with mean zero and variance $1/300$ and $299s_t^2$ is a chi-square variate with 299 degrees of freedom. It follows that, with 95 percent confidence, $\bar{z}_t$ should lie in the interval $(-.113, +.113)$ and s_t should lie in the interval $(.919, 1.079)$.

4. Standardized Error Statistics for Smoothed Estimates

t	Model of parameter evolution					
	Random walk ^a		Discrete jump ^b		Stable Markov ^c	
	$\bar{z}_t$	s_t	$\bar{z}_t$	s_t	$\bar{z}_t$	s_t
2	-.01	1.08	.32	.99	.12	2.25
3	-.01	1.14	.38	1.06	-.01	1.91
4	-.01	1.19	.48	1.01	-.02	2.02
5	-.01	1.15	.57	1.07	.03	1.81
6	-.08	1.13	.69	.97	-.08	2.72
7	-.15	1.22	.80	.93	-.10	2.71
8	-.20	1.21	.92	.93	-.07	2.85
9	-.09	1.14	1.07	.94	-.18	2.79
10	.00	1.14	1.25	1.01	-.11	2.48
11	-.06	1.11	1.48	1.05	-.19	2.53
12	-.02	1.06	1.81	.95	-.36	2.76
13	-.05	1.14	2.24	.89	-.18	2.55
14	.01	1.09	2.85	.82	.06	2.15
15	-.04	1.08	-.93	1.08	.08	1.53
16	-.02	1.22	-.69	1.00	-.20	2.06
17	.02	1.16	-.53	1.04	-.20	2.01
18	-.01	1.24	-.45	.99	-.08	2.34
19	.02	1.36	-.40	.88	-.04	2.94
20	.02	1.29	-.36	.86	.04	2.78
21	-.09	1.16	-.31	.88	.14	2.43
22	-.01	1.19	-.27	.96	.03	2.13
23	-.04	1.12	-.26	.97	-.04	1.99
24	-.02	1.15	-.14	.91	.02	2.36
25	-.03	1.15	-.22	.96	.28	2.69
26	.01	1.12	-.22	.93	-.05	2.17
27	.14	1.04	-.20	.91	-.09	2.15
28	.00	.97	-.18	.85	-.09	2.79
29	-.04	1.08	-.16	.84	-.08	2.71
30	.07	1.17	-.14	.89	.05	2.14
31	.10	1.17	-.12	1.06	-.13	1.28

^a Case I, $P = .1$.

^b Case II, $\Delta = 108$.

^c Case III, $P = .1$.

Table 4 shows that the smoothed coefficient estimates for Case I are unbiased but have a slightly larger sampling variance than that implied by $\delta^2 V_t$. This results from

ignoring the sampling error of $\hat{P}$ in computing V_t . The variances of the smoothed estimators are considerably understated by $\sigma^2 V_t$ in Case III as a consequence of failing to recognize the incomplete persistence of the p_t 's. The predictable reversion of β_t towards unity in Case III is not recognized by an estimation procedure that assumes the coefficients follow a random walk. In Case II the smoothed estimators are biased upwards in periods before the coefficient shift, and downwards in periods after the shift. The standardized bias was found most severe for short sample intervals and for small shifts. For given sample interval and shift magnitude, it is always more severe in the neighborhood of the shift point. In general, however, Table 4 indicates that the smoothed parameter estimates and estimation variances from vpr, based on the estimated $\hat{P}$, can be quite useful in tracing the true parameter trajectory through time.

5. THE STABILITY OF THE DEMAND FOR MONEY

In a recent article Khan [10] applied the BDE cusum of squares statistic to test whether the coefficients in a money demand function were stable during the twentieth century. He was unable to reject the hypothesis of stability in any of eight versions of the demand function. However, we saw in the previous section that the cusum of squares test is not as powerful as the likelihood statistic of vpr in rejecting a false null hypothesis under a variety of patterns of coefficient variation. This suggests that Khan's conclusions may be either strengthened or reversed by applying variable parameter regression to his model.

Khan's model is:

$$\Delta \ln (M_t) = \beta_{1,t} + \beta_{2,t} \Delta \ln (R_t) + \beta_{3,t} \Delta \ln (Y_t) + u_t \quad (5.1)$$

where M_t is real per capita money balances, R_t is an interest rate, and Y_t is real per capita income. While we make no criticism of this model here, we do note that, expressed in first differences, it implies an asymptotically unbounded variance on the demand for money as time goes on. β_1 is a time rate of growth coefficient because of the differencing.

Our data are taken from the volume of historical data published by the U.S. Department of Commerce [20]. The aggregate quantity of money is either the narrow money supply $M1$ (Series B109 and B110 in the foregoing volume) or the broader $M2$ (B111 and B112). Aggregate money is deflated by population (A114) and the GNP deflator (B61 and B62). The interest rate is either the commercial paper yield Rcp (B80), or the corporate bond rate Rcb (B74). Income is measured by net national product (A6). Table 5 compares the stationary estimation results of (5.1) with the corresponding results reported by Khan for his data set.

Application of the cusum of squares test to the four versions of the money demand model gave results similar to Khan's. The test statistic was never significant at the 5 percent level of confidence. In each version, however, the statistic moved away from the mean value line $(t - n)/(T - n)$ in mid-1940 and almost became significant in 1950. This feature was also present in Khan's figures.

For the reasons discussed in Section 2, a straightforward application of the likelihood statistic on the null hypothesis that $P = 0$ is not possible when the dimension of the coefficient vector is greater than one. We can, however, test whether individually estimated coefficient variances are significantly different from zero while maintaining the hypothesis that the other variances are zero. Table 6 reports the estimated values of $[P]_{t,t}$ when

5. Stationary Estimates of Demand for Money^a

Definition of money	Period ^b	Constant	$\Delta \ln(Rcp)$	$\Delta \ln(Rcb)$	$\Delta \ln(Y)$	R^2	d.w.	s.e.
$\Delta \ln(M_1)$	1916-1966	.00200 (.40)	-.0833 (2.86)		.4947 (4.71)	.360	1.43	.051
	K	.009 (1.47)	-.085 (2.86)		.332 (3.69)	.280	1.60	.049
	1916-1966	.00499 (.68)		-.3198 (3.17)	.4147 (4.04)	.381	1.53	.050
	K	.012 (2.03)		-.303 (4.61)	.208 (2.49)	.332	1.66	.046
$\Delta \ln(M_2)$	1891-1966	.01477 (2.73)	-.0697 (3.07)		.3850 (4.93)	.294	1.46	.045
	K	.017 (2.99)	-.070 (2.65)		.279 (3.46)	.183	1.63	.045
	1901-1966	.01614 (2.75)		-.2688 (2.98)	.2942 (3.50)	2.84	1.43	.046
	K	.019 (3.63)		-.256 (4.30)	.175 (2.32)	.300	1.50	.041

^a "t" values shown in parentheses.

^b Corresponding estimates from Khan [10] denoted by K.

6. Single Variance VPR Estimation of (5.1)

Definition of money	Interest rate	Growth rate			Interest elasticity			Income elasticity		
		$\hat{\rho}_{11}$	$\hat{\sigma}$	$-2\ln(\lambda)$	$\hat{\rho}_{22}$	$\hat{\sigma}$	$-2\ln(\lambda)$	$\hat{\rho}_{33}$	$\hat{\sigma}$	$-2\ln(\lambda)$
M1	Rcp	.077	.044	2.99	.0	.051	.0	3.07	.049	1.41
M1	Rcb	.043	.046	1.05	.0	.050	.0	3.17	.048	1.29
M2	Rcp	.0006	.045	.05	.761	.042	.81	.0	.045	.0
M2	Rcb	.0001	.046	.001	.0	.046	.0	.0	.046	.0

$[P]_{j,k} = 0$ for $j \neq i$ or $k \neq i$. Also shown are the estimated standard deviation $\hat{\sigma}$ of the observation errors and the likelihood statistic from the null hypothesis that $P = 0$. The evidence is quite strong that there has been no variation in the coefficients of the demand for broadly defined money ($M2$) when the bond yield (Rcb) is used as the interest rate variable. Since a chi-square distribution with one degree of freedom provides a conservative confidence level with respect to rejecting a true null hypothesis, it appears that there may have been some instability in the coefficients of the demand for $M1$ when the commercial paper rate is used.

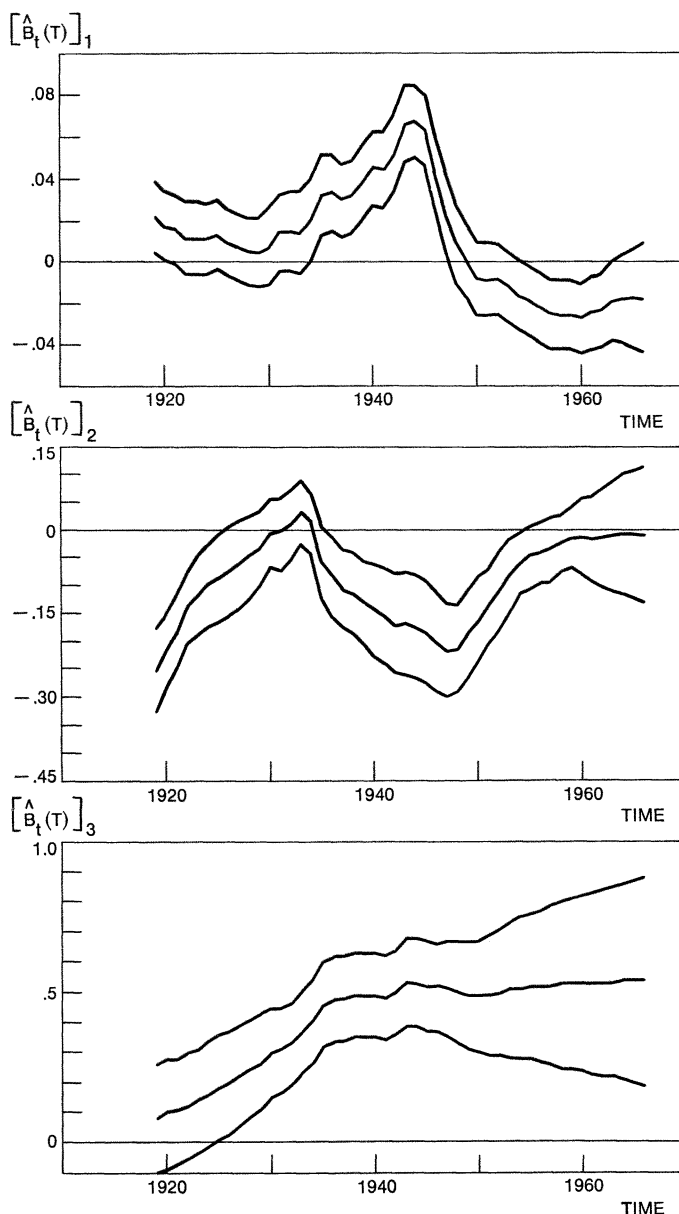
To see whether this instability was quantitatively significant, we maximized the concentrated likelihood function (2.18) with respect to the three diagonal elements of P simultaneously, and then used the estimated P to compute the smoothed coefficient estimates. Table 7 reports the estimated elements of P and the estimated observation standard deviation $\hat{\sigma}$ for each of the four versions of (5.1). Note that the variances for the growth rate and income elasticities of demand are either negligible or zero when the broadly defined money supply is used, and that the variance of the corporate bond rate elasticity of demand is zero when either definition of the money stock is used. The variances of all three coefficients are quite large, however, when narrowly defined money and the commercial paper rate are used.

7. Parameters of VPR Estimation of (5.1)

Definition of money	Interest rate	$\hat{\rho}_{11}$	$\hat{\rho}_{22}$	$\hat{\rho}_{33}$	$\hat{\sigma}$
M1	Rcp	.1982	2.6791	5.4574	.0337
M1	Rcb	.0203	.0	1.8783	.0461
M2	Rcp	.0001	.7590	.0	.0418
M2	Rcb	.0001	.0	.0	.0458

The figure shows the smoothed estimates of the coefficients of (5.1) for $M1$ and the commercial paper rate, bordered by plus and minus one standard error of estimate. The fluctuations are substantial and well-defined for the growth rate and the interest elasticity, but the income elasticity appears to have stabilized midway through the Great Depression at about .5. The estimated interest elasticity of the demand for narrowly defined money has fallen to zero in the postwar period, a phenomenon examined recently by Cagan and Schwartz [3]. The estimated growth rate of demand has turned (insignificantly) negative from the rather high value during

the Second World War. Although not exhibited here, the pattern of variation in the estimated elasticity of demand for $M2$ with respect to the commercial paper rate is almost identical to the interest elasticity estimates shown in the figure. When the narrowly defined money supply $M1$ is coupled with the bond rate rather than the commercial paper rate the estimated income elasticity and growth rate behave as in the figure.

Smoothed Coefficient Estimates for Demand for $M1$ Using Commercial Paper Rate

The striking similarity in the smoothed coefficient estimates when different data series are used suggests substantial variation in the elasticity of the demand for either definition of money with respect to the commercial paper rate. The rate of growth of demand for $M1$ seems to have changed sharply during the unusual conditions of the Depression and the Second World War, but has since returned to a level not much different from that which prevailed during the 1920s. We believe the likelihood statistics failed to detect a significant variation because the assumption of randomly walking coefficients is probably not an adequate description of the actual process of coefficient change. Specification of the demand for money appears to require a more complex model than (5.1).

6. CONCLUSIONS

This paper investigated the problem of detecting and estimating coefficient variation with two alternative methodologies, the tests of Brown, Durbin, and Evans and variable parameter regression. With the exception of quantitatively small shifts, the likelihood statistic of vpr was more powerful than the cusum of squares test in rejecting a false null hypothesis of stability. When persistence in coefficient variation is reduced, both vpr and the cusum of squares test become less powerful. Note, however, that the power of vpr may be increased in the latter case if the researcher is willing to estimate values for M and β^* in (4.8), which is more difficult and expensive than using the simple structure of (2.1). Finally, neither methodology gave misleading results when the coefficients were in fact stable.

We also considered Khan's results on the stability of the demand for money, and found the stability of different functions differ sharply. Fluctuations in the elasticity of demand with respect to the commercial paper rate have been substantial in this century, as have fluctuations in the trend growth rate and the income elasticity of demand for $M1$. Demand for $M2$ using the corporate bond rate has been quite stable. These findings are sharper than those of Khan. Khan claimed, and we agree, that the demand function was more stable when the bond rate was used, but he could not distinguish between the two alternative definitions of the money stock. Our results indicate that the demand function for broadly defined money is more stable.

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